Buddy Collins

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Education

University of Colorado Boulder | M.S. in Network Engineering — Expected May 2028
- Graduate Certificate in Project Management — GPA: 3.7/4.0
- Focus: Cybersecurity, LAN/WAN Architecture, Network Reliability, Cloud Networking.

University of Colorado Boulder | B.A. in Geography, Geospatial Information Systems (GIS) — May 2020

Professional Experience

National Center for Atmospheric Research (NCAR) | Boulder, CO

Network Engineer — July 2020 – Present
- Architect, deploy, and maintain research-grade WAN/LAN and DWDM optical transport infrastructure, supporting core routing and out-of-band (OOB) management networks across NCAR facilities and 31 Front Range GigaPoP (FRGP) partner organizations.
- Configure, troubleshoot, and maintain multi-vendor optical transport systems, primarily ADVA/ADTRAN, including ROADMs, coherent optics, transponders, and optical amplifiers to ensure resilient network connectivity.
- Direct field teams and conduct precision optical testing using OTDRs, optical spectrum analyzers (OSAs), and power meters across 720+ miles of fiber to isolate transport faults, execute emergency restorations, and maintain physical-layer integrity.
- Lead multi-vendor configuration management, routine patching, and infrastructure documentation, including network architectures, fiber routes, and disaster recovery procedures, to strengthen network security, support operational compliance, and incident response.
- Developed 10+ interactive GIS web applications and enterprise geodatabases to monitor physical infrastructure while mentoring 8 student interns on diagnostic tools, network operations, and documentation workflows.

Student Assistant — July 2019 – July 2020
- Supported senior engineers during critical network outages, performed emergency remote-hands operations, assisted with fiber deployments and inventory management, and developed custom ArcGIS Online applications to map and document physical network infrastructure.

Technical Projects

Network Automation & Provisioning CLI Tool (Python, Netmiko, SSHv2, GNS3, Linux)
- Engineered an automated Python CLI tool using argparse and Netmiko to remotely provision interfaces, configure loopbacks, verify IP reachability, and backup running configurations over SSHv2.
- Implemented automated timing buffers and error handling to replace manual CLI interaction, targeting configuration overhead across a simulated multi-device environment.

Enterprise L2/L3 Topology and Protocol Diagnostics (Cisco IOS, GNS3, Wireshark, 802.1Q, STP)
- Designed multi-switch virtual networks implementing 802.1Q VLAN trunking, sub-interface Inter-VLAN routing ("router-on-a-stick"), and Spanning Tree Protocol (STP) loop prevention.
- Deployed multi-segment DHCP relay agents and local DNS mappings, validating packet-level transport (DORA exchanges, ARP tables, ICMP type/code structures) via Wireshark captures.

Technical Skills & Competencies

Optical Networking & Infrastructure
- DWDM, ROADMs, Coherent Optics, Optical Transponders/Amplifiers, ADVA/Adtran, Fiber Optics, Fiber Characterization, OTDR, OSA, Optical Power Meters, Visual Fault Locators, Transport Troubleshooting

Routing, Switching & Protocols
- Cisco IOS, Juniper, MikroTik, WAN/LAN & Core Routing Architecture, OOB Management, BGP, OSPF, VLANs, 802.1Q Trunking, STP, NAT, ACLs, GNS3, TCP/IP, UDP, IPv4/IPv6, ARP, ICMP, DNS, DHCP, SSH, Wireshark, Packet Analysis

Programming & Automation
- Python, C++, JavaScript, HTML, CSS, R, GitHub, Linux/Bash automation.

Geospatial & Data Analytics
- ArcGIS Suite (Pro, Online, ArcPy), QGIS, GDAL, OGR, GeoPandas, Pandas, GeoJSON, KMZ, Leaflet, Fiona.

Affiliations & Leadership

SCinet (Super Computing Network Team) WAN Team Member (SC23) — Fall 2023
- Collaborated with international engineers to design, deploy, and maintain the world's fastest temporary research network supporting Supercomputing conference infrastructure.

FRGP (Front Range Giga-Pop) Wide Area Network Engineer — 2021 – Present
- Design, deploy, document and maintain non-profit research area infrastructure in direct collaboration with senior network engineering leads.

WINS (Women in Networking at SC) Digital Engagement Lead — 2023 – 2025
- Managed web presence and outreach programs to expand visibility for diversity and inclusion initiatives within the high-performance computing and SCinet communities.